

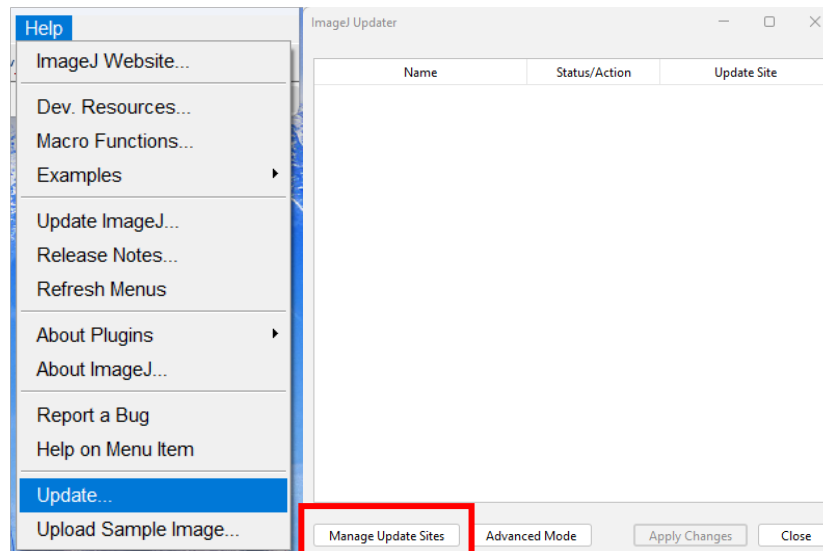
Complete User Guide for DH-PSF analysis with DHPSFU

This user guide is divided into two parts:

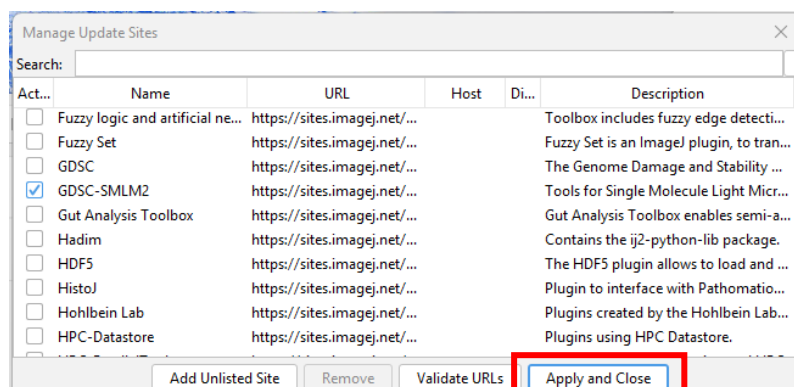
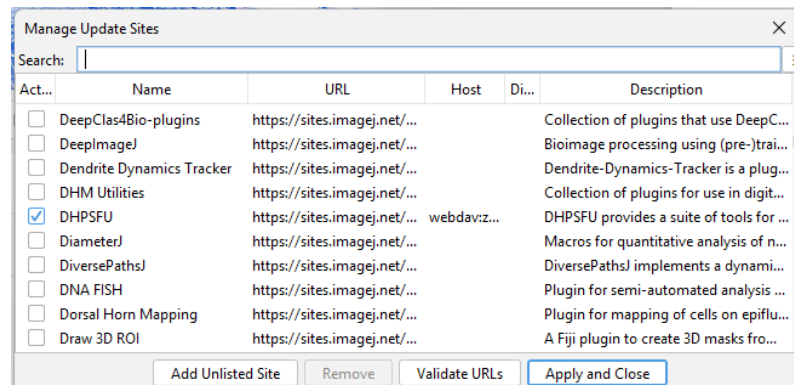
- Installation of the DHPSFU plugin.
- Analysis procedure for DH imaging data.

Part 1: Installation of the DHPSFU plugin in FIJI

1. Download and install ImageJ (Fiji) from <http://imagej.net/software/fiji/downloads>
2. Launch Fiji and go to **Help > Update...** to open the ImageJ Updater.



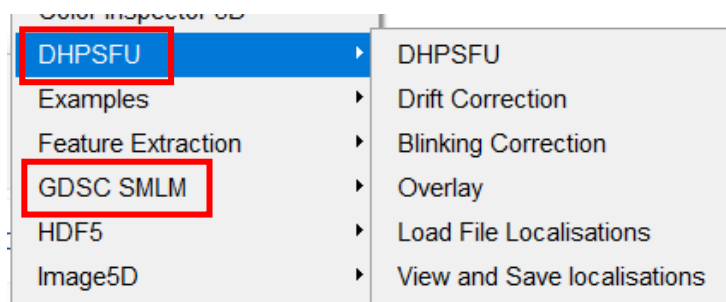
3. Click on **Manage Update Sites**, tick both **DHPSFU** and **GDSC SMLM2**.



4. A list of plugins to be installed will appear in the ImageJ Updater, click **Apply Changes** to install.



5. Once complete, you should see **DHPSFU** and **GDSC SMLM** under the **plugin** tab.



Analysis procedure for DH imaging data.

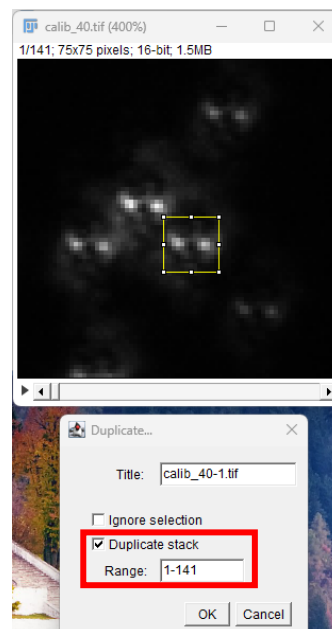
The analysis consists of two main steps:

- Localisation detection with **GDSC SMLM**.
- 3D coordinate calculation with **DHPSFU**.

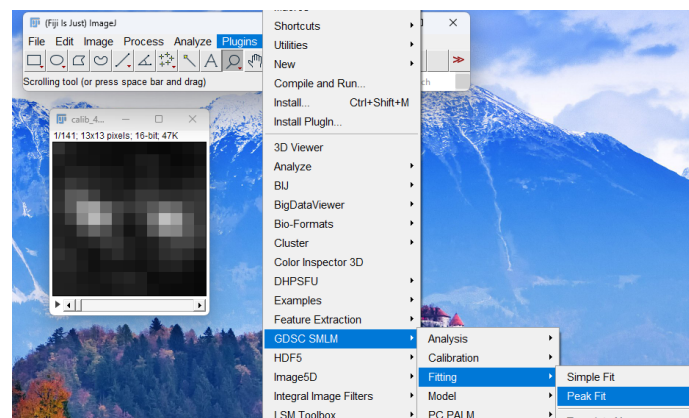
The example provided uses resPAINT data of the apical plasma membrane of a Jurkat T cell.

Step1: Localisation detection with GDSC SMLM

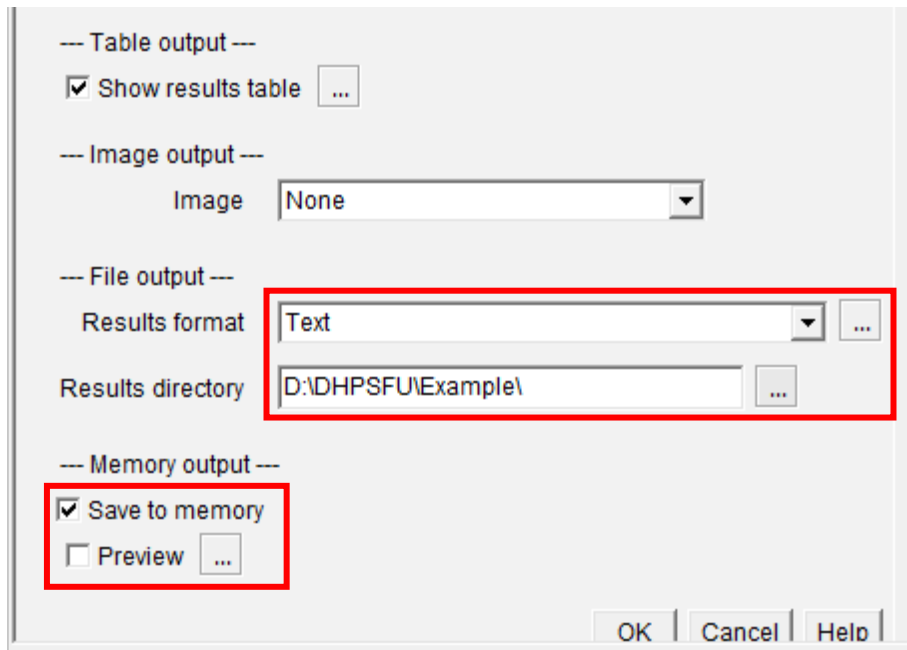
1. Load the Peak Fit template: Go to **Plugin > GDSC SMLM > Fitting > Template Manager** and select **Load custom templates**. Choose the example folder and load **PFparameters.txt**
2. Open the calibration image stack **calib_40.tif**.
3. Use the **Rectangle** tool in ImageJ to select a small region of interest (ROI) around a DH-PSF you intend to use for analysis.
4. Press **Ctrl + Shift + D** to crop and duplicate the ROI.



5. Run Peak Fit: go to **Plugin > GDSC SMLM > Fitting > Peak Fit**, select the **PFparameters** template.



6. Results format, select Text, and specify your Results directory to where you would like the result to be saved. Tick Save to memory. This generates a file `calib_40-1.tif.results.xls`.

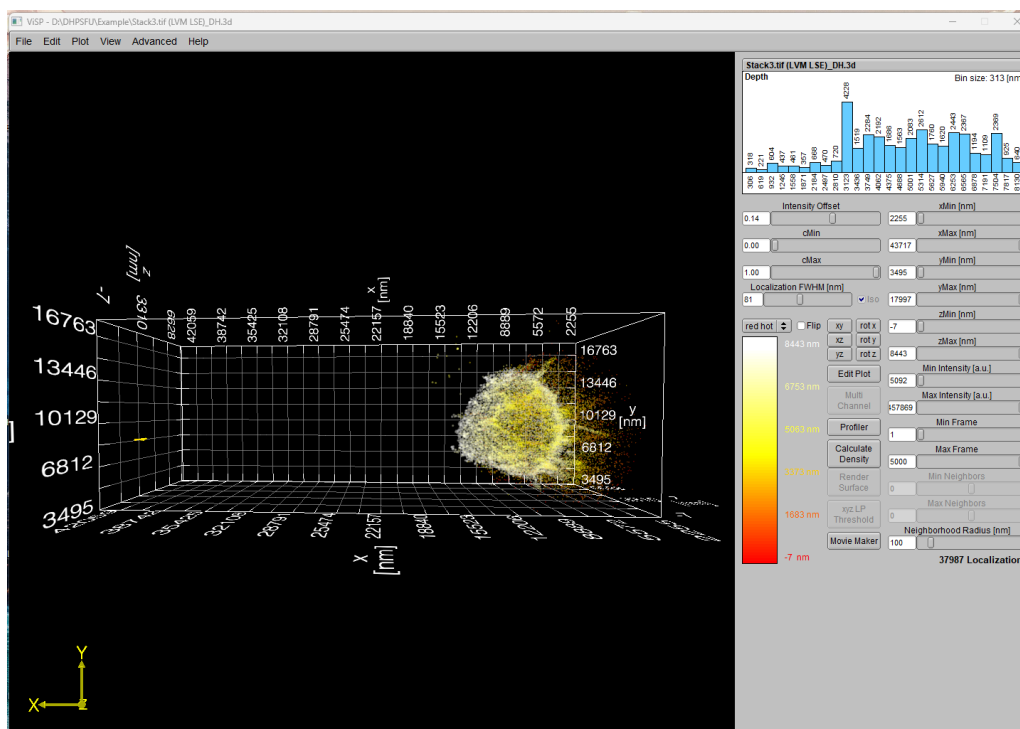


7. Peak Fit will detect 282 localizations with the provided parameters.
8. Open the image stack `Stack3.tif`, which is an example DH imaging stack.
9. Repeat the Peak Fit with the same template. Set Precision to 30. A file `Stack3.tif.results.xls` will be saved in your results directory.

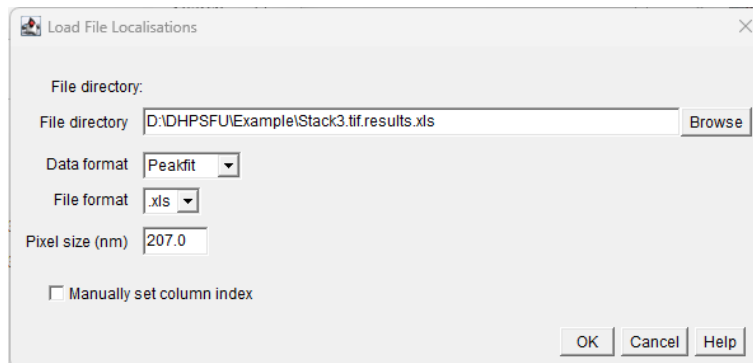
Step2: 3D coordinate calculation by DHPSFU

- Go to **Plugin > DHPSFU > DHPSFU**, select **calib_40-1.tif (LVM LSE)** as **Calibration file** and **Stack3.tif (LVM LSE)** as **Data file**. Ensure both Peak Fit results are stored in ImageJ memory. ImageJ memory. Apply the following parameters and specify the save directory.

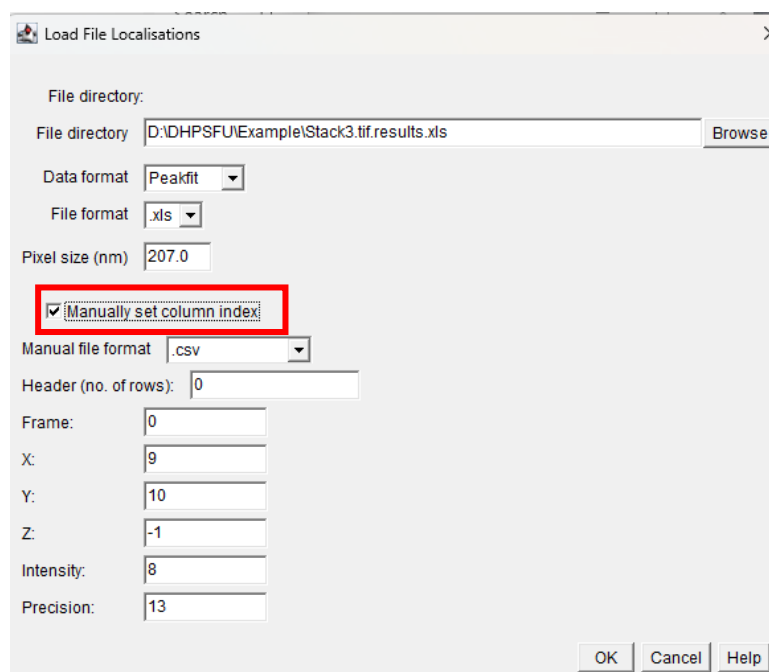
- The following output files will be saved:
 - Stack3.tif (LVM LSE)_DH.3d**: 3D localisations
 - Stack3.tif (LVM LSE)_DH.3dlp**: with calculated precision
(Both are .tsv-formatted text files.)
- You can view these files in ViSP (Available at: <https://www.nature.com/articles/nmeth.2566>).



4. You can also load PF localisation files or 3D localisation files into ImageJ memory: go to **Plugin > DHPSFU > Load File Localisations** and browse for the data file you would like to load.



5. If using a different localization format, choose **Manually set column index** to map input columns (index starts at 0).



6. You are all set! Let us know if you encounter any issues.